

- 9 (a) The maximum kinetic energy of the photoelectrons is dependent only on the frequency, but not on the intensity of the incident radiation.

In the particulate theory of light, electromagnetic radiation is made up of discrete quanta of energy known as photons. Each photon has energy $E_{\text{photon}} = hf$ where h is the Planck constant and f is the frequency. Emission of a photoelectron is only possible when a single photon is absorbed by an electron on the surface of the metal in a one-to-one interaction.

The maximum kinetic energy of the photoelectron is given by $K.E._{\text{max}} = hf - \Phi$ where Φ is the work function of the metal. The equation shows that the maximum kinetic energy is a function of the photon's energy and hence frequency for the same metal with constant Φ .

Increasing the intensity of light only increases the rate of incident photons on the metal and has no effect on E_{photon} which is dependent only on frequency. Therefore, there is no effect of intensity on the maximum kinetic energy which provides evidence for the particulate nature of electromagnetic radiation.

OR

There is a threshold frequency below which there is no emission of photoelectrons regardless of the intensity of the incident radiation.

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The maximum kinetic energy of the photoelectron is given by $K.E._{\text{max}} = hf - \Phi$ where Φ is the work function of the metal. The emission of a photoelectron is only possible if the maximum kinetic energy is greater than zero hence $hf \geq \Phi$. If the photon energy is less than Φ i.e. $f < f_0$ where $f_0 = \frac{\Phi}{h}$ is the threshold frequency, there will be no emission of photoelectrons.

Increasing the intensity of light only increases the rate of incident photons on the metal and has no effect on E_{photon} which is dependent only on frequency. Therefore, there is no effect of intensity on the threshold frequency which provides evidence for the particulate nature of electromagnetic radiation.

(b) (i)
$$\Phi = \frac{hc}{\lambda_0}$$
$$\lambda_0 = \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{(4.33)(1.60 \times 10^{-19})}$$
$$= 2.8709 \times 10^{-7} = 287 \text{ nm}$$

(ii)
$$E_{K, \max} = \frac{1}{2} m v_{\max}^2 = \frac{hc}{\lambda} - \Phi$$

$$v_{\max} = \sqrt{\frac{2}{m} \left(\frac{hc}{\lambda} - \Phi \right)}$$

$$= \sqrt{\frac{2}{9.11 \times 10^{-31}} \left(\frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{210 \times 10^{-9}} - (4.33)(1.60 \times 10^{-19}) \right)}$$

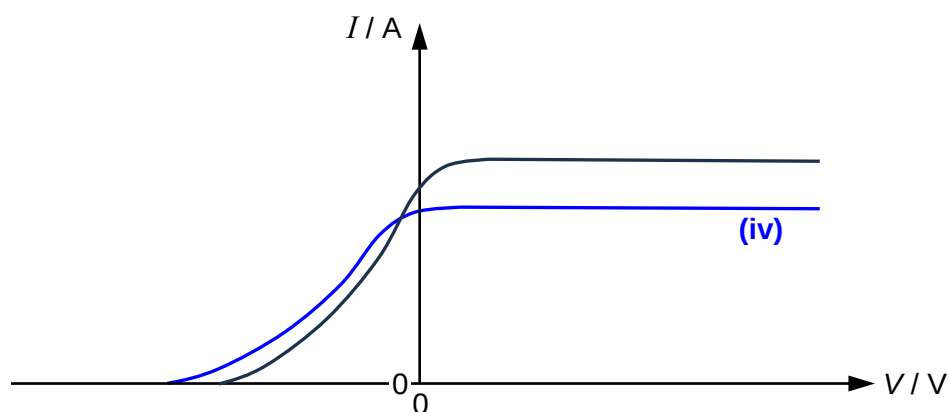
$$= 7.4725 \times 10^5 = 7.47 \times 10^5 \text{ m s}^{-1}$$

- (iii) When $V = 0 \text{ V}$, there is no electric force on the emitted photoelectrons and they travel at a constant speed along a straight line perpendicular to the collector plate.

When $V = +3.0 \text{ V}$, the electric force, $F_e = eE = \frac{eV}{d} = \frac{3e}{d}$, is the resultant force on the photoelectron and it acts in the same direction as its velocity.

Since the resultant electric force is constant, acceleration, $a = \frac{3e}{m_e d}$, is constant, and the speed of the emitted photoelectron increases at a constant rate as it moves along a straight line perpendicular to the collector plate.

- (iv)



Shorter wavelength, higher photon energy, higher maximum kinetic energy, hence larger stopping potential.

Since intensity $i = \frac{hc}{\lambda A} \left(\frac{dN_p}{dt} \right)$ remains the same, the rate of incident photons $\frac{dN_p}{dt}$

decreases with a decrease in wavelength or an increase in photon energy. Hence the saturated photocurrent decreases.

- (c) (i) The electrons behave as waves with de Broglie's wavelength λ which is related to its momentum p by $\lambda = \frac{h}{p}$. The graphite film, having atoms that are regularly spaced at distances comparable to the wavelength of the electrons, acts as a diffraction grating resulting in a diffraction pattern of concentric rings when the electrons pass through the graphite film.

The maximum and minimum intensities of the concentric rings are a result of electron waves undergoing constructive and destructive interference similar to how light waves interfere.

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(ii)
$$E_k = \frac{1}{2}mv^2 = \frac{1}{2} \frac{(mv)^2}{m} = \frac{p^2}{2m} = \frac{(h/\lambda)^2}{2m} = \frac{h^2}{2m\lambda^2}$$

increase in kinetic energy = decrease in electric potential energy

$$\frac{h^2}{2m\lambda^2} - 0 = q\Delta V$$

$$\lambda = \frac{h}{\sqrt{2mq\Delta V}}$$

$$= \frac{6.63 \times 10^{-34}}{\sqrt{2(9.11 \times 10^{-31})(1.60 \times 10^{-19})(5.0 \times 10^3)}}$$

$$= 1.7366 \times 10^{-11} = 1.74 \times 10^{-11} \text{ m}$$

- (iii) According to Rayleigh criterion, the limiting angle of resolution $\theta_{\min} \approx \frac{\lambda}{b}$ where b is the size of the aperture. For the electrons, $\theta_{\min}$ is approximately four orders of magnitude smaller ($\theta_{\min} \propto \lambda$) than for visible light.

Hence, an electron microscope has a much higher resolving power compared to an optical microscope allowing it to resolve structures at the atomic scale which the optical microscope is unable to.